

Local Area Network

- Gigabit Ethernet (Wired LAN)
- Bluetooth (Wireless LAN)
- Connecting LAN (Devices)
- Virtual LAN

Learning Outcome

- **DIFFERENTIATE ACCESS CONTROL FOR LAN AND WAN I.E. CSMA/CD, TDMA (C4)**

EXPLAIN TYPES OF LAN TECHNOLOGY I.E GIGABIT ETHERNET, BLUETOOTH (C2)

EXPLAIN TYPES OF WAN TECHNOLOGY I.E CELLULAR, SATELITE (C2)

EXPLAIN METHOD OF VIRTUAL LAN (C2)



What is LAN?

A **local area network (LAN)** is a network that interconnects computers and network devices over a relatively **short distance**. A networked office building, home or school usually contains a single LAN, though sometimes one building will contain a few small LANs (perhaps one per room), and occasionally a LAN will span a group of nearby buildings.

LANs are also typically **owned, controlled, and managed** by a single person or organization.

LAN can be generally divided into two medium; **wired and wireless**.

De facto wired LAN connectivity technologies usually are **Ethernet** and **Token Ring**. As with wireless LAN, **Wi-Fi and Bluetooth** are the most popular technologies.



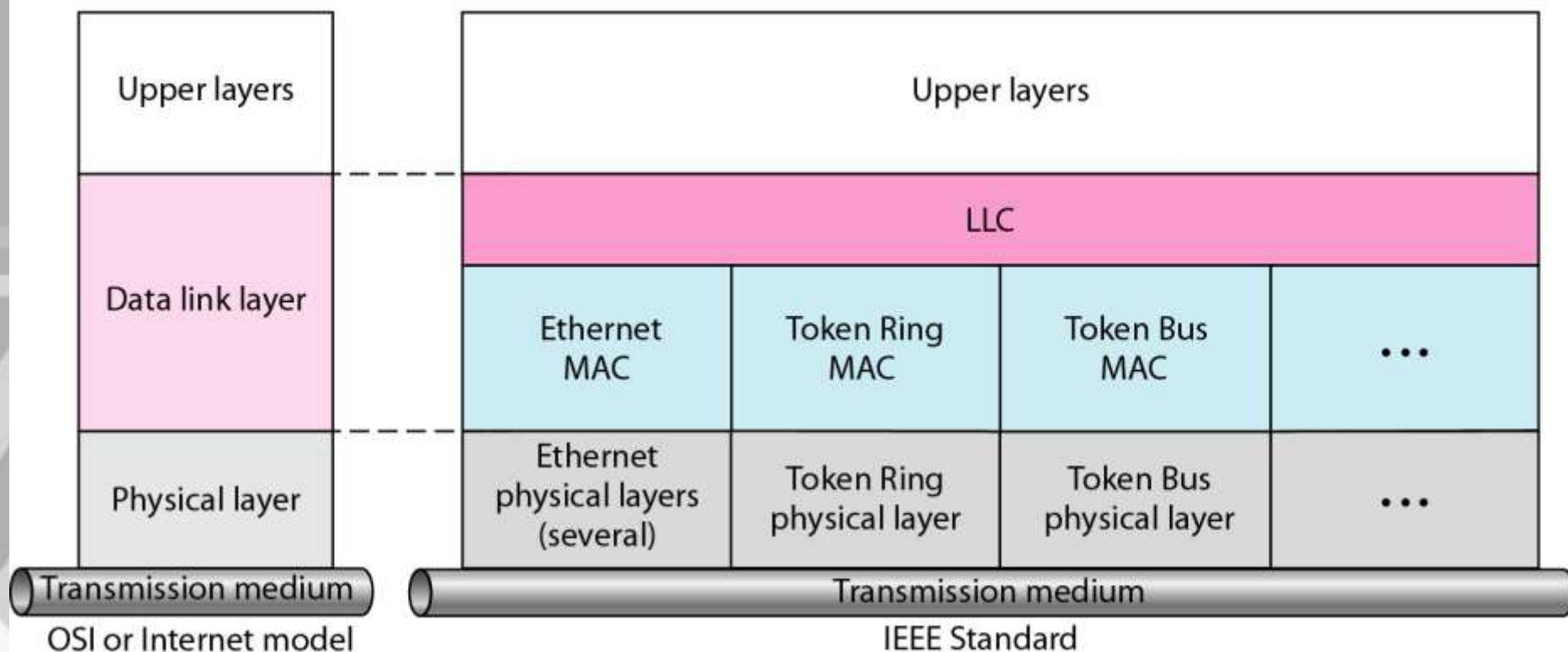
LAN Protocols

In 1985, the Computer Society of the IEEE started a project, called Project 802, to set standards to enable intercommunication among equipment from a variety of manufacturers. Project 802 is a way of specifying functions of the physical layer and the data link layer of major LAN protocols.

IEEE Standards for LANs

LLC: Logical link control

MAC: Media access control





Ethernet ...

- Most widely used LAN standard
- The original Ethernet was created in 1976 at Xerox's Palo Alto Research Center (PARC). Since then, it has gone through four generations (next slide)
- It is a IEEE 802.3 standard
- Used CSMA/CD access control

CSMA (Carrier Sense Multiple Access)

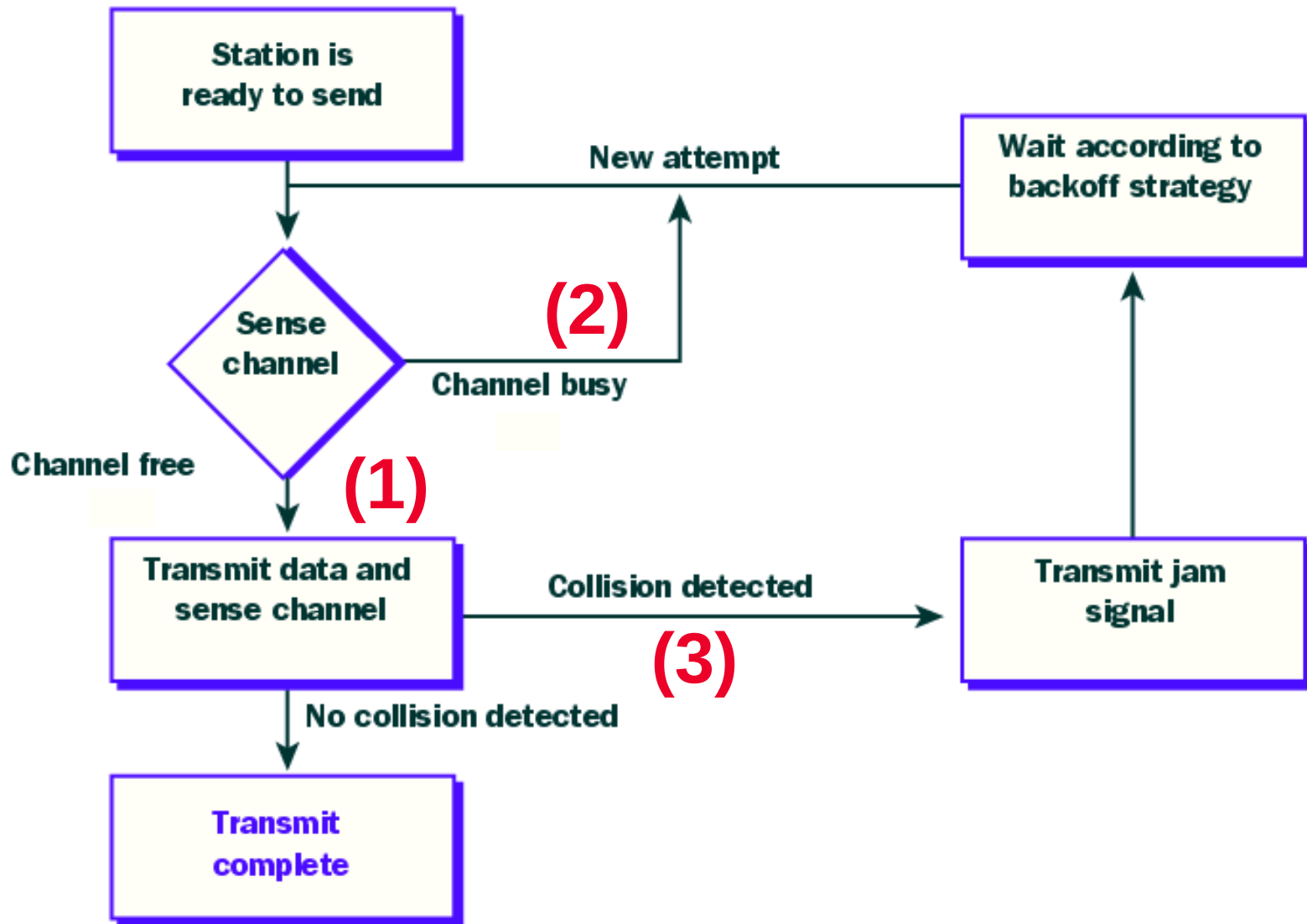
- stations soon know transmission has started
- so first listen for clear medium (carrier sense)
- if medium idle, transmit
- if two stations start at the same instant, collision
 - wait reasonable time
 - if no ACK then retransmit

CSMA/CD Description

CSMA/CD rules:

1. if medium idle, transmit
2. if busy, listen for idle, then transmit
3. if collision detected, transmit jam signal and then stop transmission, wait random time then retry

CSMA/CD



Ethernet Evolution

Figure 13.3 *Ethernet evolution through four generations*

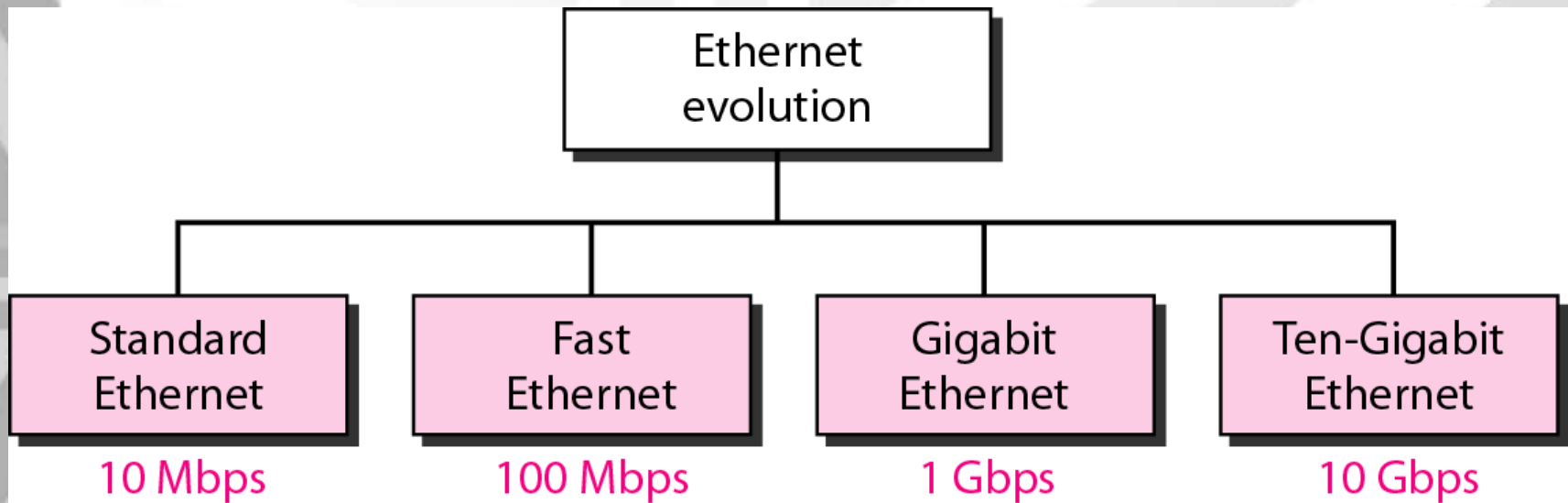


Figure 13.6 Example of an Ethernet address in hexadecimal notation



06 : 01 : 02 : 01 : 2C : 4B

6 bytes = 12 hex digits = 48 bits

```
C:\WINDOWS\system32\cmd.exe
Windows IP Configuration

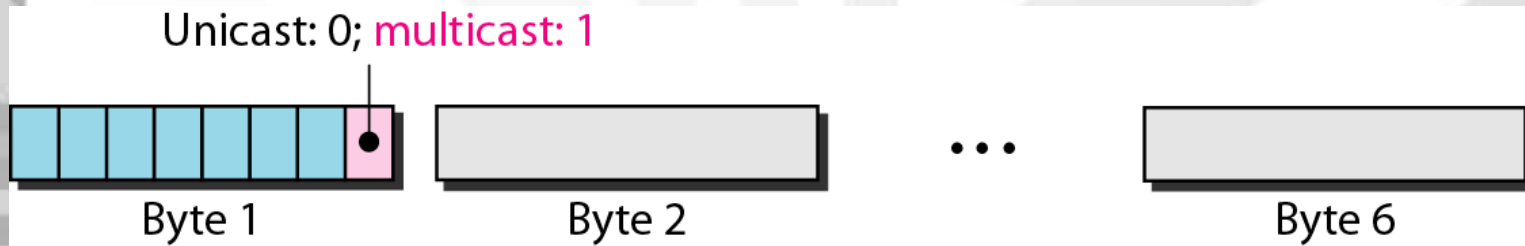
Host Name . . . . . : dell-idawaty
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : upm.edu.my

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . : upm.edu.my
Description . . . . . : Intel(R) 82566DM-2 Gigabit Network C
onnection
Physical Address. . . . . : 00-21-9B-33-B1-84
Dhcp Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IP Address. . . . . : 172.16.61.136
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 172.16.61.254
DHCP Server . . . . . : 172.16.240.20
DNS Servers . . . . . : 172.16.240.250
                        172.16.240.251
Lease Obtained. . . . . : Thursday, September 15, 2011 7:23:13
```

Ethernet
Address

Figure 13.7 *Unicast and multicast addresses*



Note

The least significant bit of the first byte defines the type of address.
If the bit is 0, the address is unicast;
otherwise, it is multicast.

Note

The broadcast destination address is a special case of the multicast address in which all bits are 1s.



Define the type of the following destination addresses:

- a. 4A:30:10:21:10:1A
- b. 47:20:1B:2E:08:EE
- c. FF:FF:FF:FF:FF:FF

Solution

To find the type of the address, we need to look at the second hexadecimal digit from the left. If it is even, the address is unicast. If it is odd, the address is multicast. If all digits are F's, the address is broadcast. Therefore, we have the following:

- a. This is a unicast address because A in binary is 1010.*
- b. This is a multicast address because 7 in binary is 0111.*
- c. This is a broadcast address because all digits are F's.*



Hexadecimal to binary

1. Split the hex number into individual values.
2. Convert each hex value into its decimal equivalent.
3. Next, convert each decimal digit into binary, making sure to write four digits for each value.
4. Combine all four digits to make one binary number.

Example - hex 28 to binary

2 = decimal 2 8 = decimal 8

2 = binary 0010 8 = binary 1000

Result - 00101000

Example - hex FC to binary

F = decimal 15 C = decimal 12

15 = binary 1111 12 = binary 1100

Result - 11111100

<https://www.bbc.co.uk/bitesize/guides/zd88jty/revision/5>



Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F



Show how the address *47:20:1B:2E:08:EE* is sent out on line.

Solution

The address is sent left-to-right, byte by byte; for each byte, it is sent right-to-left, bit by bit, as shown below:

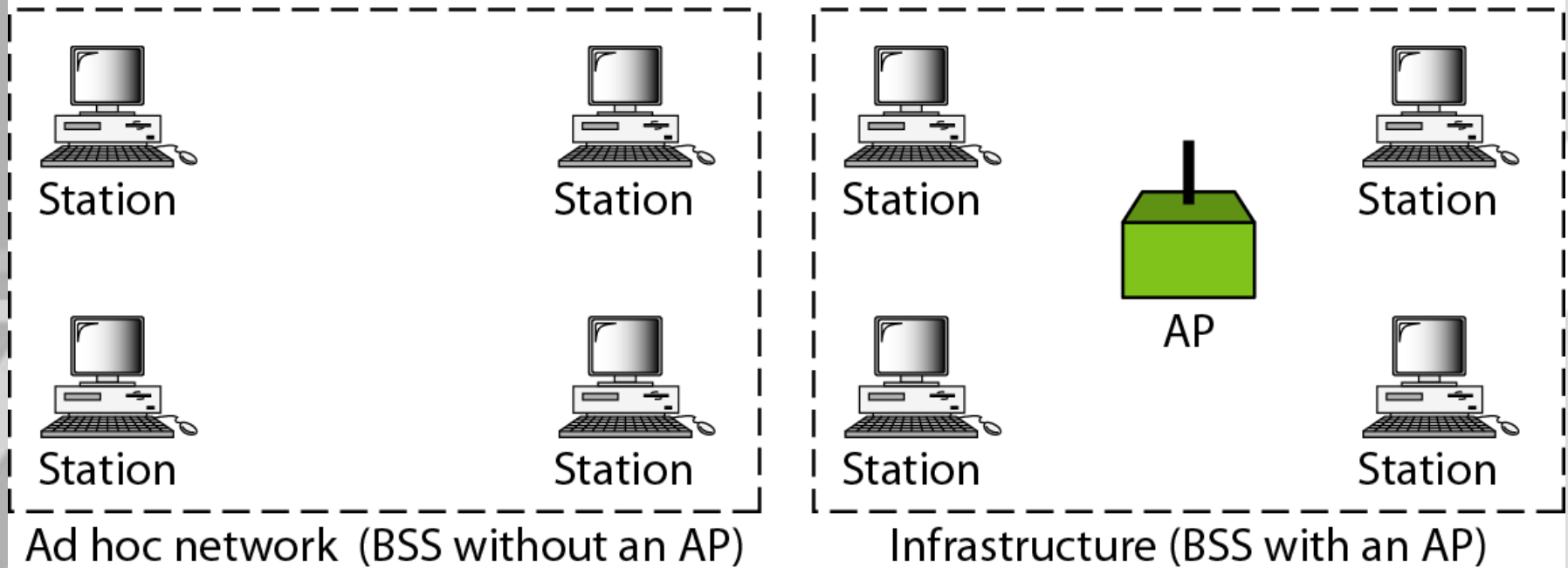
← 11100010 00000100 11011000 01110100 00010000 01110111

WIRELESS LAN

Figure 14.1 Basic service sets (BSSs) (Wireless LAN)

BSS: Basic service set

AP: Access point



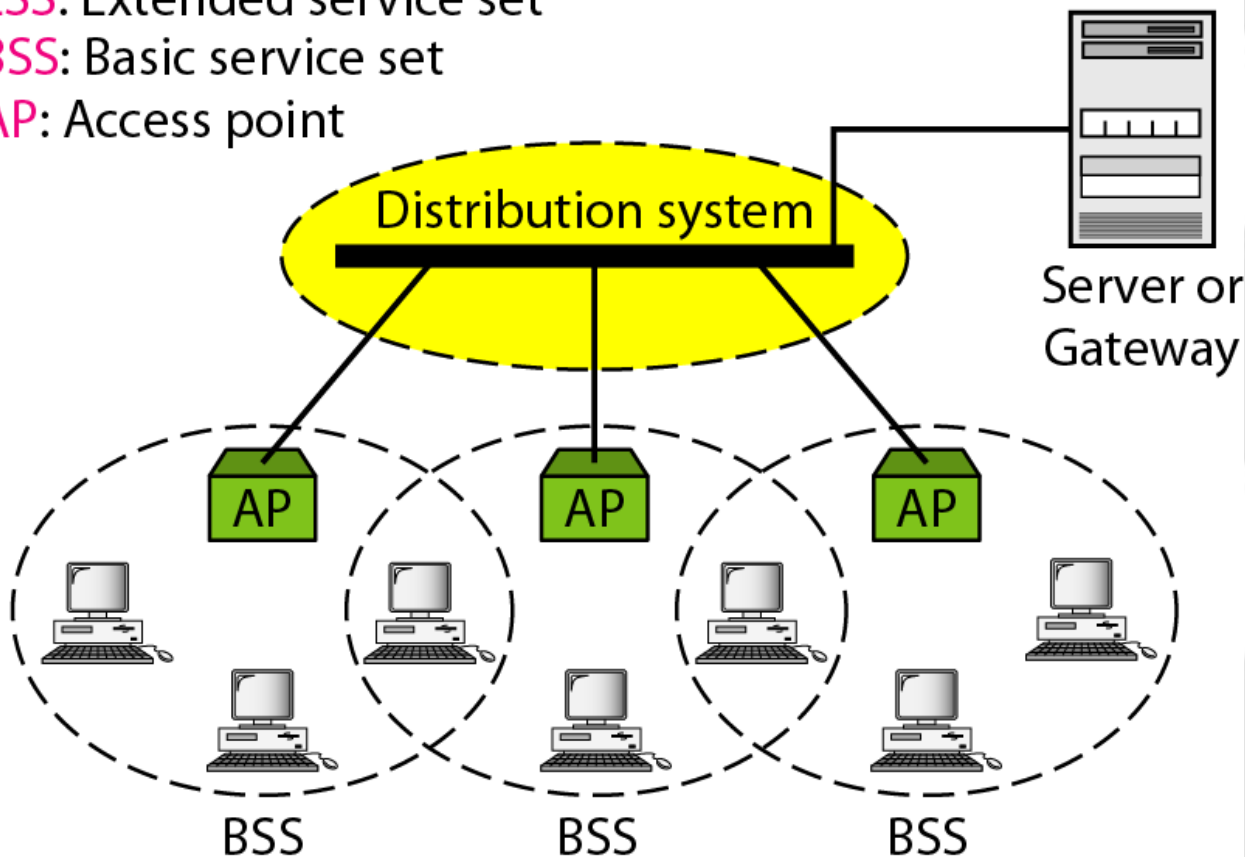
A BSS without an AP is called an ad hoc network;
a BSS with an AP is called an infrastructure network.

Figure 14.2 *Extended service sets (ESSs)*

ESS: Extended service set

BSS: Basic service set

AP: Access point



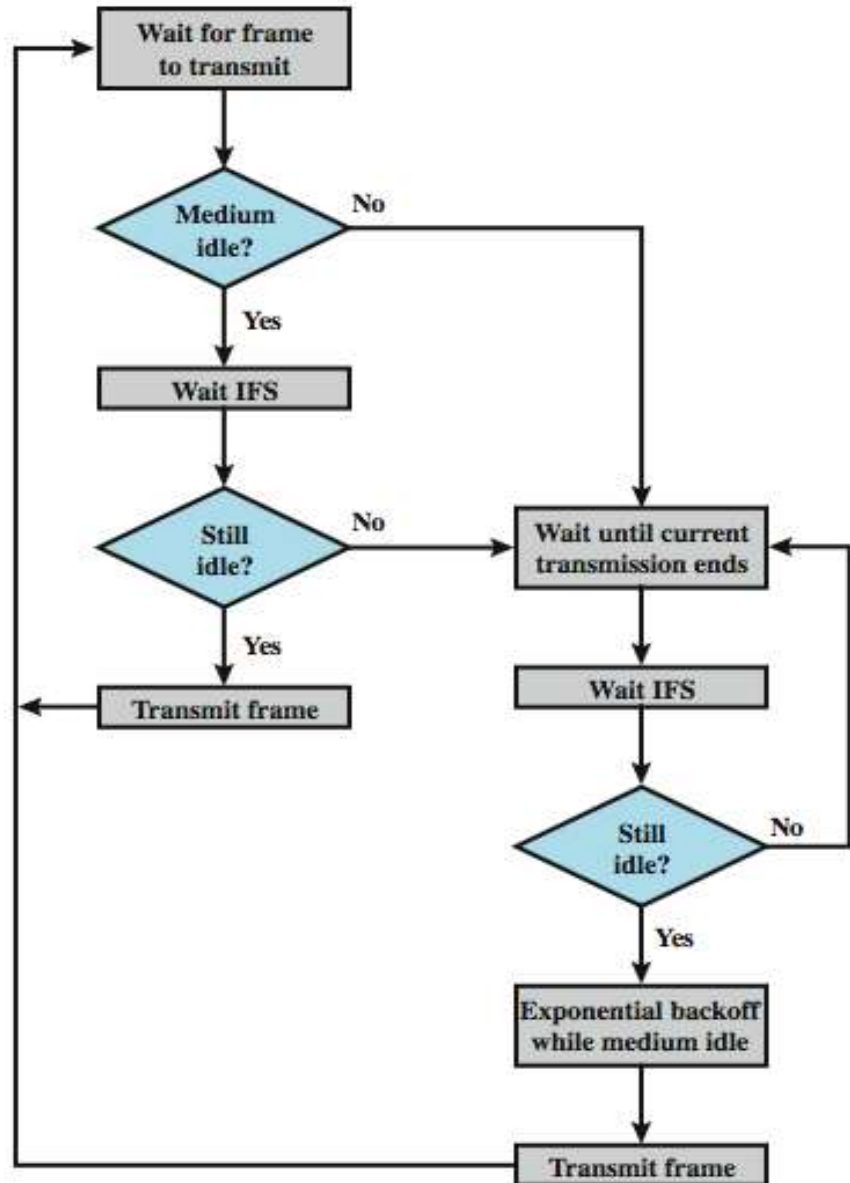


CSMA/CA

IEEE 802.11

MAC Protocol

IFS : Inter Frame Spacing





CSMA/CA Description

CSMA/CA rules:

1. if medium idle, wait for some IFS time, then sense the channel
 1. If busy – go to step 2
 2. If still idle – transmit a frame
2. if busy, wait until transmission end and then wait for some IFS time, then sense the channel again
 1. If busy – go to Step 2
 2. If idle – backoff/wait for some random time and transmit a frame

Bluetooth

IEEE has defined the specifications for a wireless LAN, called IEEE 802.11, which covers the physical and data link layers.

Bluetooth is a wireless LAN technology designed to connect devices of different functions such as telephones, notebooks, computers, cameras, printers, coffee makers, and so on. A Bluetooth LAN is an ad hoc network, which means that the network is formed spontaneously.



What is Bluetooth?

- Proprietary short range wireless technology.
 - Intended to create PAN or Personal Area Network.
 - Operate range of 10 meters
 - 1mW transmission power.
- Operate in the free unlicensed ISM band from 2400-2480 MHz.
 - The transceiver utilizes frequency hopping to reduce interference and fading.
 - Can support both data (asynchronous) and voice (synchronous) communications, with total bandwidth of 1 Mb/sec
- Provides multiple levels of security
 - Spread spectrum frequency hopping
 - Data encoding
 - Authentication
 - Data encryption



Basic usage of Bluetooth

- As cable replacement.
- Voice and data access points
 - Notebook PC to mobile phone
 - Notebook or PDA companion to LAN, email, Internet via shared access point (e.g. conference room)
- Ad-hoc/temporary networking
 - Up to 8 devices in a single Bluetooth network.
- Advantages :Low cost wireless solution for short range distance.
- Global technology specification – an universal radio interface in 2.4 GHz ISM frequency band.
- Stationary and mobile environments.
- No setup required – always in the background (dynamic configuration)



Bluetooth: Piconet

Bluetooth devices can interact with one or more devices in several different ways.

- Only 2 devices → point-to-point piconet
- More than 2 devices → point-to-multipoint piconet.

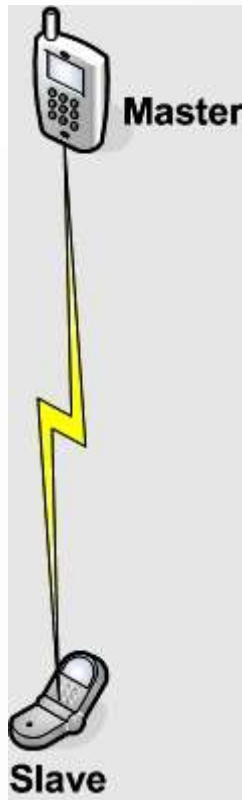
A piconet is any such Bluetooth network with one master and one or more slaves. Master controls entire piconet.

Maximum devices in a piconet:-

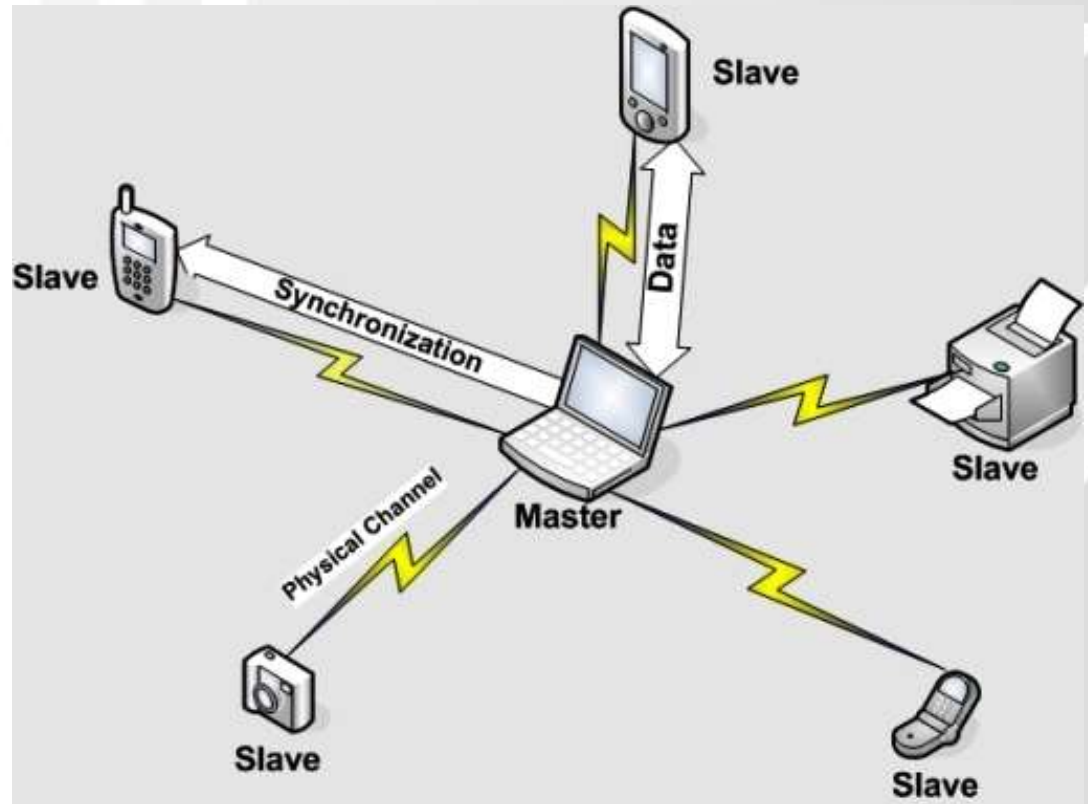
- 1 master
- 7 active slaves
- 255 standby slaves

Devices can participate in more than 1 piconet.

Bluetooth: Piconet

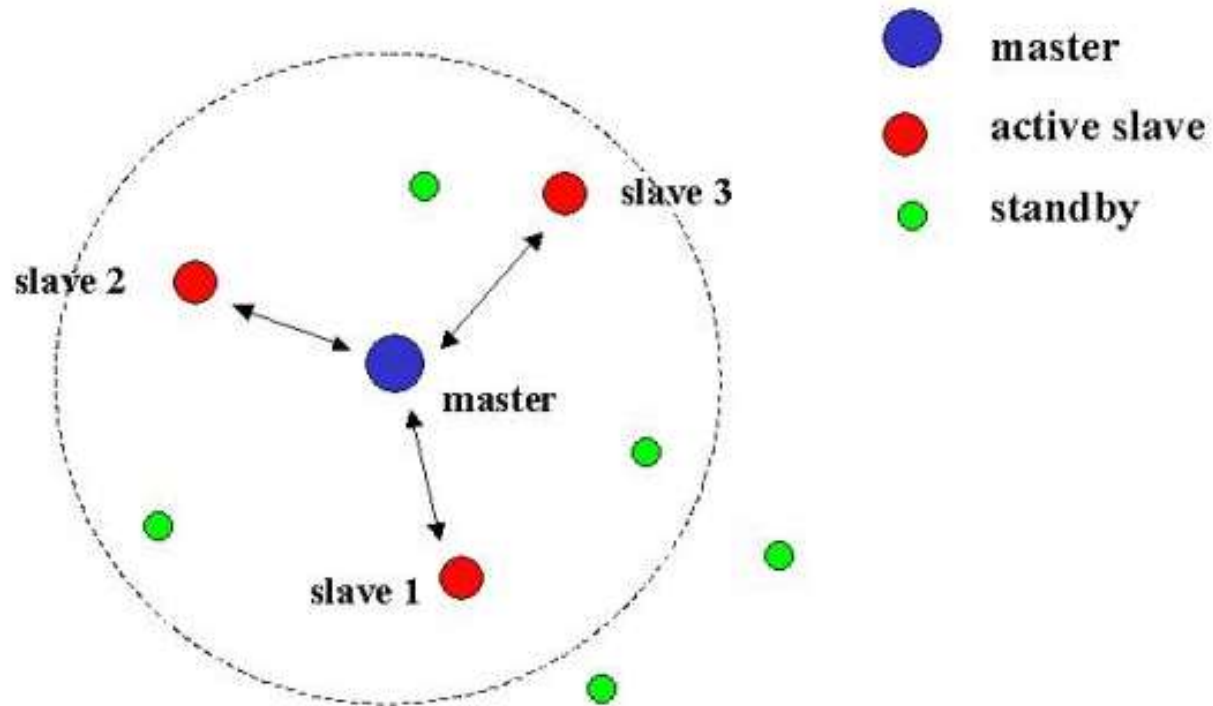


point-to-point



point-to-multipoint

Bluetooth: Piconet



Bluetooth: Scatternet

- A group of piconets with overlapping areas of coverage → scatternet
- Each piconet is identified by different frequency hopping sequence
- Slave to 1 piconet can be master to another
- A device can only be Master to a single piconet.

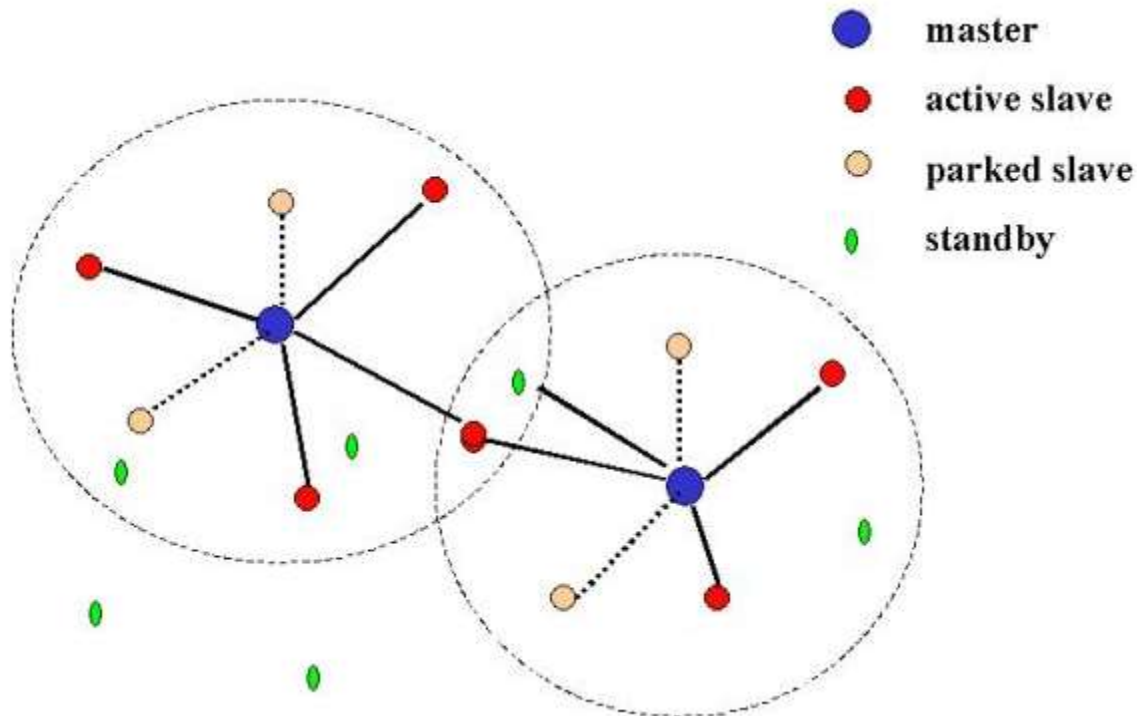
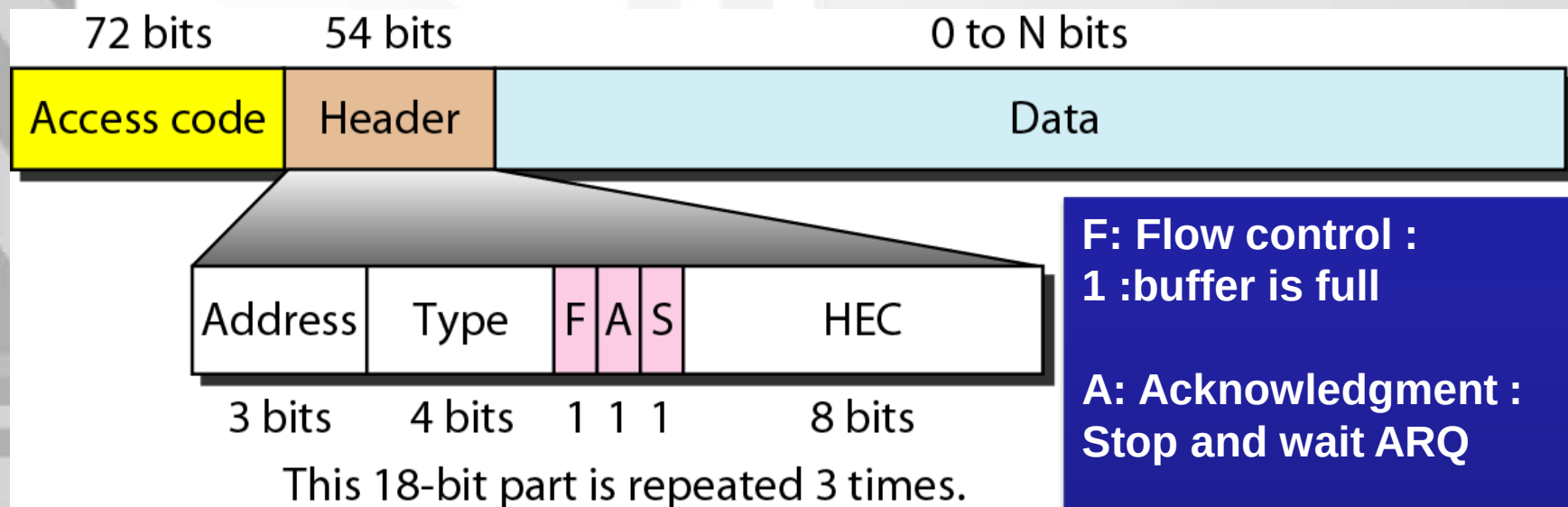


Figure 14.24 *Frame format types*



Address : For 7 secondaries

Header has 3 identical 18 bits sections to form forward error correction (for header) using HEC

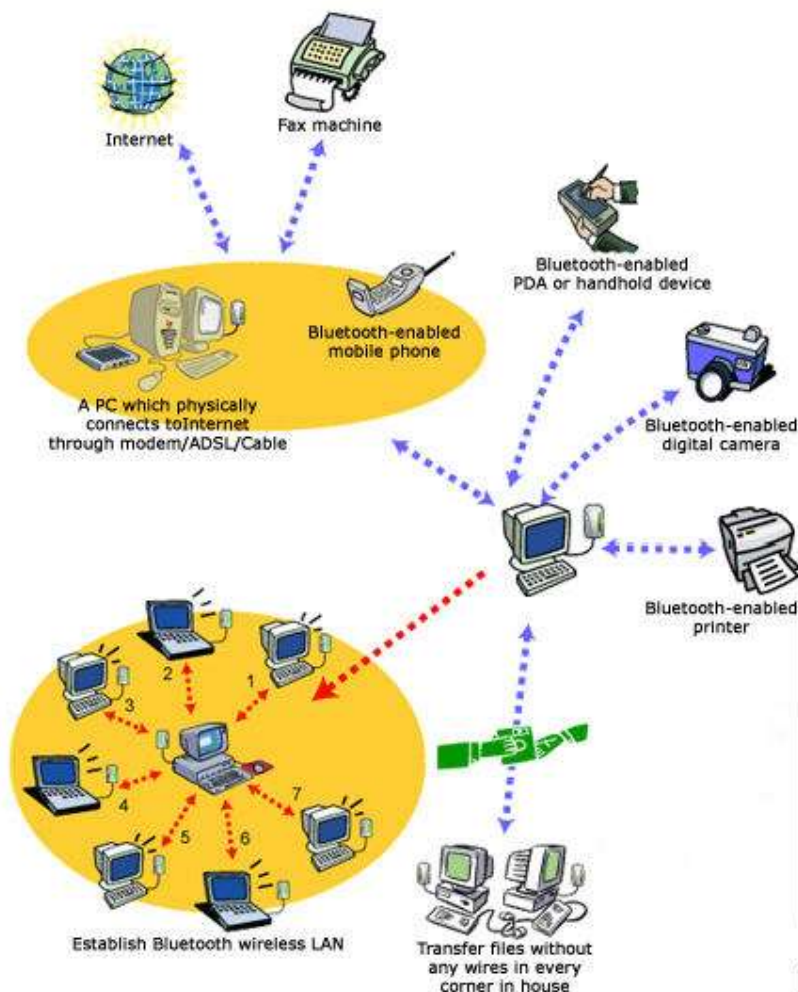
**F: Flow control :
1 :buffer is full**

**A: Acknowledgment :
Stop and wait ARQ**

**S: Sequence number
for Stop and Wait ARQ**

**HEC: Header Error
Correction**

Bluetooth: Scatternet Example



- One computer is master and accessing different slave Bluetooth devices, such as Bluetooth printer, Bluetooth digital camera, etc.
- The master computer is also connected to another master computer.
- The other master computer is slave to the 'central' master computer.
- But, this slave computer is master to his Bluetooth networking slave computers.